

Original Article

He-Chan Pian inhibits the metastasis of non-small cell lung cancer via the miR-205-5p-mediated regulation of the GREM1/Rap1 signaling pathway

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ABSTRACT

Background: He-Chan Pian (HCP), a traditional Chinese medicinal formula, shows promising efficacy for the treatment of lung cancer.

Purpose: Gremlin (GREM1) plays an important role in gastrointestinal tumor metastasis; however, little is known about its role in lung cancer. We determined the mechanism underlying the protective effect of HCP against metastasis in a mouse model of non-small cell lung cancer (NSCLC) and demonstrated the role of GREM1.

Methods: Ultra-high performance liquid chromatography-mass spectrometry (UPLC-MS) was used to analyze the herbal components and metabolites from the serum of HCP-treated mice. The tumor, liver, and kidney were examined histologically, and the antitumor effects and toxicity of HCP were evaluated. Levels of epithelial-mesenchymal transition (EMT)-associated transcription factors were measured using western blotting in tumors from five groups (i.e., model, HCP [L], HCP [M], HCP [H], and positive control [cisplatin, DDP]). Differentially expressed proteins and genes were identified using protein chip and sequencing analyzes, respectively. Short hairpin RNAs and overexpression plasmids were introduced into cells to evaluate the effects of GREM1. To evaluate proliferation, migration, and invasion, the expression levels of proteins involved in the Rap1 pathway and EMT were measured *in vitro*. Xenograft tumors with overexpression-GREM1 (OE-GREM1) in A549 cells were examined for cell proliferation. A dual-luciferase assay was performed to verify the direct interaction of GREM1 with miR-205-5p in lung cancer.

Results: Thirty-six ingredients and bioactive constituents detected in the serum of HCP-treated mice were identified as the key compounds involved in the inhibition of tumor growth. Animal experiments revealed that HCP significantly decreased tumor volumes and had no adverse effects on the liver or kidney or side effects. GREM1 upregulation was closely related to tumor metastasis and was regulated by miR-205-5p, as confirmed using a dual-luciferase reporter assay. OE-GREM1 promoted A549 cell migration and invasion, promoted EMT, and increased the expression of Rap1 pathway intermediaries, whereas shGREM1 had the opposite effects.

Abbreviations: BMP, bone morphogenetic protein; CCK-8, cell counting kit-8; DDP, Cisplatin; EMT, epithelial-mesenchymal transition; FasL, Fas ligand; GREM1, Gremlin; HCP, He-Chan Pian; IL-9, Interleukin-9; IL-15, Interleukin-15; NSCLC, non-small cell lung cancer; NS, normal saline; OE-GREM1, overexpression-GREM1; SDF-1 α , stromal cell-derived factor-1 alpha; shRNA, short hairpin RNA; TARC, Thymus and activation-regulated chemokine; TCM, traditional Chinese medicine; TSLP, thymic stromal lymphopoietin; UPLC-MS, ultra-high performance liquid chromatography-mass spectrometry; WT, wild-type.

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Furthermore, the effects of OE-GREM1 on proliferation in the A549 xenograft mouse model were attenuated, although HCP has an inhibitory effect on tumors.

Conclusion: Our results suggest that HCP contributes to the inhibition of NSCLC metastasis via the Gremlin/Rap1 signaling pathway regulated by miR-205-5p.

Introduction

Non-small cell lung cancer (NSCLC) is one of the most detrimental malignant tumors, with the highest morbidity and mortality rates worldwide (Kundu et al., 2018). The treatment options for NSCLC include surgical resection, chemotherapy, radiotherapy, targeted therapy, and immunotherapy (Hirsch et al., 2017). Traditional Chinese medicine (TCM) is increasingly used as an adjunctive therapy for lung cancer (Li et al., 2018). Professor Zhou Daihan, a renowned TCM practitioner in China, has used TCM to treat various cancers for more than 50 years and has developed many proprietary Chinese medicines. In particular, he developed He-Chan Pian (HCP) based on accumulated clinical experience for the treatment of primary bronchial lung cancer and lung metastatic cancer. HCP is used to treat NSCLC with deficiencies in qi and yin, phlegm, blood stasis, and toxin stagnation type III and IV, and it has achieved a specific curative effect (Xi et al., 2011; Zhou et al., 1986).

The *GREM1* gene was first discovered in *Xenopus laevis* in 1988. The secreted glycosylation protein encoded by this gene can bind to bone morphogenetic protein (BMP) and act as an antagonist (Nolan and Thompson, 2014). Recently, a number of studies have reported that *GREM1* contributes substantially to tumor development and aggressiveness. *GREM1* has a paracrine/autocrine role in tumor neo-vascularization and mediates pro-angiogenic and pro-inflammatory activity simultaneously (Corsini et al., 2014; Mitola et al., 2010). Moreover, *GREM1* is closely linked to cell proliferation and branching morphogenesis during lung development and kidney organogenesis (Michos et al., 2004). However, studies of the role of *GREM1* in the invasion and metastasis of lung cancer are lacking.

Tumor metastasis is an important factor affecting treatment and prognosis; however, the process is poorly understood. Adhesion, degradation, and migration are the three stages of tumor cell invasion and metastasis. When adhesion between tumor cells is destroyed, tumor cells separate from the primary tumor and adhere to the extracellular matrix (ECM). To complete the metastasis process, cancer cells dissolve the ECM and basement membrane, enter adjacent tissues, intrude lymphatic vessels, and invade the blood and metastatic organs (Morifuji et al., 2014). HCP demonstrates substantial clinical benefits for patients with lung cancer. However, the molecular mechanisms underlying its antitumor effect are not well characterized. In this study, the effects of HCP and the role of *GREM1* in NSCLC metastasis were evaluated.

Materials and methods

Animal experiment

Forty, 5-week-old, specific pathogen-free BALB/c nude mice were obtained from Gem Pharmatech Co., Ltd., Jiangsu, China [License No. SCXK (Yue) 2018-0008]. The animal experiments were approved by the Experimental Animal Ethics Committee of the Guangzhou University of Chinese Medicine (No. 20200609007). A549 NSCLC cells were purchased from the Cell Bank of the Shanghai Institute of Biological Sciences, and the density was adjusted to 5.0×10^7 /ml. The cells and the basement membrane Matrigel were mixed at a ratio of 1:1. The A549 cells (0.2 mL) were injected subcutaneously into the right armpit of each mouse. On day 3, tumors were observed under the skin of nude mice, and the model was successfully established. HCP was purchased from Guangzhou Baiyunshan Zhongyi Pharmaceutical Company Limited. The recommended dose of HCP is 2.22 g, administered every 8 h for humans

of approximately 70 kg (adult). The dosage for mice was 1.37 g/kg according to the dosage conversion (Reagan-Shaw, 2008), which is the clinically equivalent dose, and this was used as the low dose. The ratio of dosages for the low: medium: high-dose groups was 1:2:4, namely, HCP (L) group (1.37 g/kg), HCP(M) group (2.74 g/kg), and HCP(H) group (5.48 g/kg), respectively. Forty mice were randomly divided into five groups, a model group (normal saline [NS] 0.2 ml/10 g via gavage), HCP (L) group (0.137 g/ml), HCP (M) group (0.274 g/ml), HCP (H) group (0.548 g/ml) (0.2 ml/10 g, gavage), and a positive control group (DDP, Cisplatin 0.2 ml/10 g, 2 mg/kg, via intraperitoneal injection every other day). Each group was medicated continuously for 21 days and then anesthetized by ethyl ether inhalation and euthanized with cervical dislocation.

UPLC-MS

On the 21st day, six mice from the model and HCP (H) groups were euthanized, and the peripheral blood was collected through orbital bleeding. Serum samples were prepared and analyzed using LC/MS. Briefly, LC-MS/MS was performed using an Agilent UPLC 1290 system (Thermo Fisher Scientific, Bremen, Germany) with a UPLC BEH C18 column (1.7 μ m, 2.1*100mm; Waters, Milford, MA, USA). The Q Exactive Focus mass spectrometer was used to collect primary and secondary mass spectrometry data with the fullScan-DDMS2 function under control software (Xcalibur; Thermo Fisher Scientific).

Tumor volume, weight, and inhibitory ratio

Every 3 days, body weights were recorded, and tumor lengths (mm) (a) and widths (mm) (b) were measured using a sliding caliper. The tumor volume was calculated using the formula $V (\text{mm}^3) = 1/2ab^2$. Tumor inhibition (%) = (average tumor weight in the model group-average tumor weight in the treatment group) /average tumor weight in the model group $\times 100$.

Histology assays of the liver, kidney, and tumor

The liver and spleen indexes were calculated using the following formula: liver and spleen index = liver and spleen weight/body weight $\times 10$. Tumors, liver, and kidneys were extracted, wet-weighted, and fixed in 4% paraformaldehyde solution for pathological sectioning. The thickness of paraffin sections was 4 μ m. Sections were stained with hematoxylin & eosin (H&E). Histological morphology was viewed using light microscopy (400 $\times$), and images were obtained.

Protein chip assays

Protein chip assays were conducted on the RayBio platform using G Series Mouse Cytokine Antibody Array 4000 (RayBio Technology, Guangzhou, China), and data were analyzed using GSM-CAA-4000 (RayBio Technology, Guangzhou, China). The assays were used to analyze the tumors from the model group and the HCP (H) group.

Transcriptome and small RNA sequencing

Tumor tissue mRNAs (transcripts) for the model and HCP (H) group were analyzed using mRNA-seq. RNA was analyzed using RNA-Seq at the Beijing Genomics Institute, China. Six cDNA libraries were constructed and used as the input material for RNA sample preparation.

Gene ontology functional annotations (<http://geneontology.org/page/go-database>) and Kyoto Encyclopedia of genes and genomes pathway analyzes (<http://www.genome.jp/kegg>) were performed.

Plasmid construction and lentivirus transfection

Short hairpin RNAs (shRNAs), including shGREM1 and a negative control (shControl), were constructed by Obio Technology Corp., Ltd. (Shanghai, China). The construction and packaging of OE-GREM1 and empty vectors as controls were performed by Obio Technology Corp., Ltd. Lipofectamine 2000 (Invitrogen, Waltham, MA, USA) was used for cell transfection for 72 h.

Cell counting kit-8 (CCK-8) assay

Briefly, 96-well plates were inoculated with 2×10^3 cells/well. Thereafter, 10 μ l of CCK-8 (MedChemExpress, Shanghai, China) solution was added to each well and incubated for 2 h. Absorbance at 450 nm was measured at 0, 24, 48, 72, and 96 h.

Wound-healing assay

Cells were grown to 70–80% confluence in 6-well plates, and a wound was created by dragging a plastic pipette tip across the cell surface. The cells were washed three times with phosphate-buffered saline to remove cellular debris and incubated with fresh culture medium containing 10% fetal bovine serum (FBS) at 37 °C for 24 h. Images were obtained at 0, 12, and 24 h. The wound area was measured using ImageJ (v1.53a, Wayne Rasband, National Institutes of Health, Bethesda, MD, USA).

Transwell invasion assay

Cells at a concentration of 2×10^6 cells/ml and Matrigel were mixed at a ratio of 50:1. Cells with Matrigel were resuspended in 200 μ l of F-12K containing 10% FBS and added to the upper chamber, whereas 700 μ l of F-12K containing 10% FBS was placed in the lower chamber. After incubation for 12 h, a cotton swab was used to remove the remaining matrix gel and cells in the upper chamber. Cells were fixed with 4% paraformaldehyde and stained with 0.1% crystal violet.

Western blot analysis

Cells were homogenized in lysis buffer. The protein in the lysate was quantified using the BCA Protein Assay Kit. The samples were separated by sodium dodecyl sulfate-polyacrylamide gel electrophoresis and transferred to polyvinylidene fluoride blotting membranes (Millipore, Billerica, MA, USA). The membranes were blocked with 5% bovine serum albumin for 1 h and then incubated with primary antibodies for 16 h. Primary antibodies against vimentin (1:500; CST5741, USA), Snail (1:1000; CST3879, USA), E-cadherin (1:1000; CST3195, USA), N-cadherin (1:1000; CST13116, USA), Rap 1 (1:1000; CST2399, USA), phospho-erk1/2 (1:1000; CST4370, USA), erk1/2 (1:1000; CST4695, USA), phospho-c-Raf (1:1000; CST9421, USA), and c-Raf (1:1000; CST9422, USA) were used to probe the membranes. After they were washed three times with TBST (10 min each), the membranes were incubated with secondary antibodies for 1 h. A chemiluminescence system was used to detect and visualize protein bands (Bio-Rad, Hercules, CA, USA).

Dual-luciferase reporter assay

A wild-type (WT) or mutant (mut) GREM1 fragment and the luciferase reporter gene of the pMIR-REPORT plasmid were constructed by Obio Technology Corp., Ltd. 293T cells were transfected with empty pMIR-REPORT-NC, pMIR-GREM1-WT, or pMIR-GREM1-MUT. The cells were then co-transfected with miR-205-5P using Lipofectamine 2000

(E1910; Beijing, China). At 48 h after transfection, the relative luciferase activity of firefly and *Renilla* was measured using the dual-luciferase reporter system. Results are shown as the firefly/*Renilla* luminescence ratio as a percentage of the untreated control.

Xenograft experiments

Male BALB/c nude mice aged 7–8 weeks were used. Different cells [A549 (control), vector, and OE-GREM1 cells] (5×10^6 cells/100 μ l/mouse) were subcutaneously injected into the mice for model establishment. The animal experiments were approved by the Experimental Animal Ethics Committee of the Guangzhou University of Chinese Medicine (No. 20201011054). Xenografts appeared in the subcutaneous tissue on the 18th day after injection. Mice were randomly split into five groups ($n = 8$ /group): control, vector, OE-GREM1, OE-GREM1 (100 μ l, NS) (OE-GREM1 + NS), and OE-GREM1 (HCP, 5.48 g/kg) (OE-GREM1 + HCP). NS and HCP were administered every day for 30 days. Body weight and tumor volume were recorded every 3 days. At the end of the experiment, the mice were euthanized with ether, and tumors were removed.

Statistical analysis

The data were analyzed using GraphPad Prism (v 8.0, GraphPad Software Inc., La Jolla, CA, USA). If the data were normally distributed and the variances were homogeneous, one-way ANOVA was used. The Tukey test was used for multiple comparisons. If the variances were not uniform, Dunnett's T3 test was used. Values for two groups were compared using the independent sample *t*-test, and values of $p < 0.05$ were considered significant.

Results

Analysis of HCP in the blood serum

UPLC-MS was used to explore the serum of HCP-treated lung cancer model mice. The retention time, UV, and MS spectra were compared with those for reference samples, and 36 major components/metabolites of HCP were identified from the serum of treated mice, as shown in Fig. 1 and Table 1. Ten monomers of the 36 major components and metabolites had antitumor functions: cearoin, sarracenin, picein, resistogenin, 14-deoxyandrographolide, celastrol, dehydroabietic acid, cinnamic acid, apigenin-7-O-glucoside, and palmitic acid. These findings provided insight into the factors to the antitumor effects of HCP.

HCP inhibits lung cancer tumor growth in vitro with no liver and kidney toxicity

On day 7, the tumor volumes were lower in the HCP (M), HCP (H), and DDP groups (Fig. 2A). On day 14, the tumor volumes were significantly lower in the HCP (M) and HCP (H) groups than in other groups, and there was a trend toward a decreased tumor volume as the dose increased (Fig. 2B). On day 21, the tumor weight and volume decreased significantly in the HCP (H) group (Fig. 2C, D). During 21 days of treatment, weight tended to increase in all groups (Fig. 2E). The body weight change in the HCP (H) group was significantly lower than that in the Model group after 21 days of treatment (Fig. 2F). The post-treatment tumor inhibition ratios (IRs) were 6.51%, 14.58%, 29.38%, and 36.40% in the HCP (L), HCP (M), HCP (H), and DDP groups, respectively. After 21 days of treatment, there were no significant differences in the liver or kidney indices among groups (Fig. 2G, H).

The dissection process was smooth, and the tumor could be completely separated. The tumor tissues and surrounding tissues could be clearly distinguished. The texture of the tumor was relatively hard, and the boundary was clear. The volumes in the model group were higher than those in the other groups. After sectioning and staining with

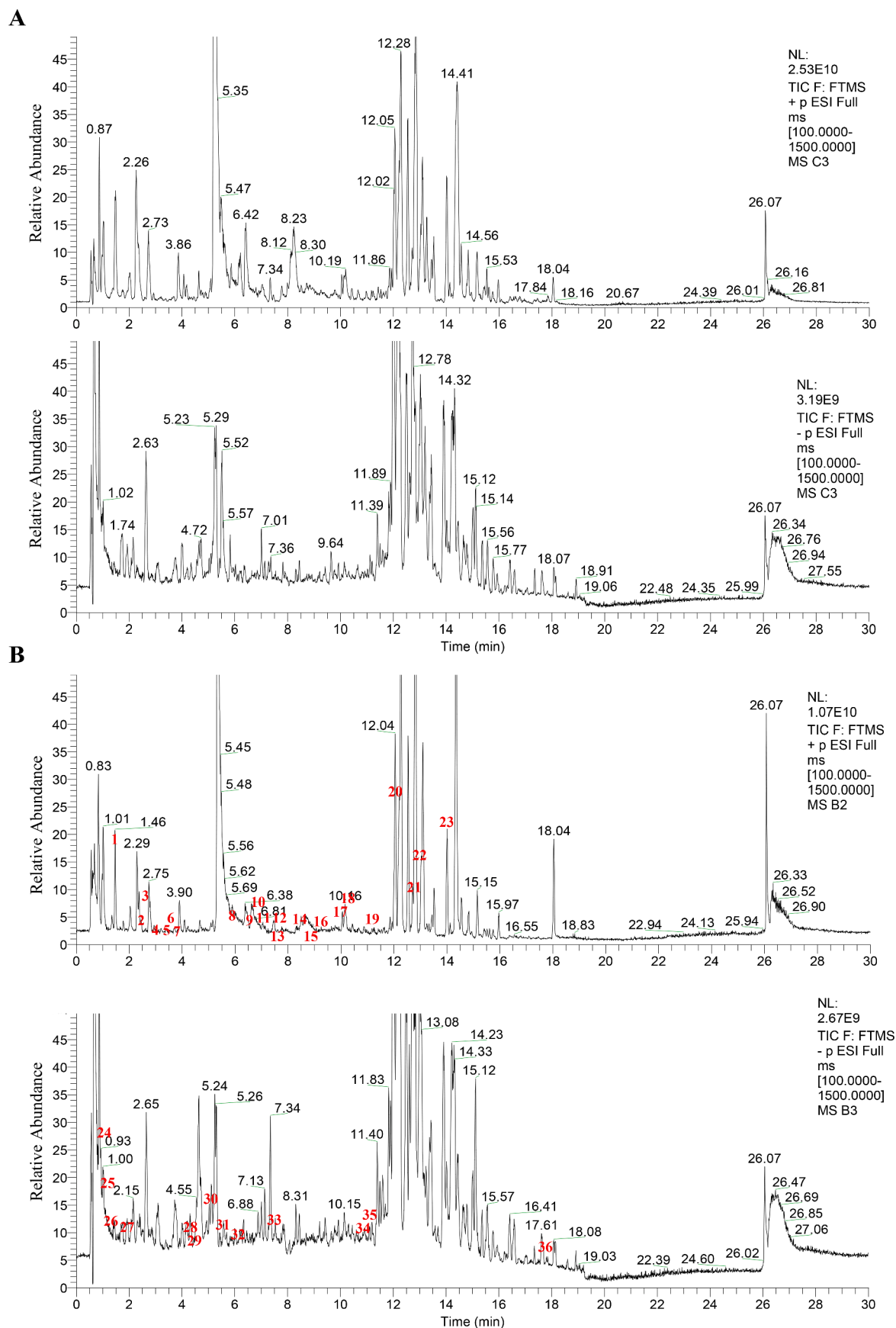


Fig. 1. Overview of 36 components of HCP detected in the blood serum using UPLC-MS [Blank serum ion map (A) and HCP-containing serum ion map (B)].

Table 1

Thirty-six components/metabolite were found in blood serum of HCP by UPLC-MS compared with model group.

Peak no.	Components/Metabolite name	Formula	InChIKey	RT (min)	Quasi-Molecular Ion (m/z)	Cal.m/z
1	N-Methyl-2-pyrrolidone	C5H9NO	SECXISVLQFMRJM-UHFFFAOYSA-N	1.46	100.075 [M+H] ⁺	100.0755
2	PHENYLALANINE	C9H11NO2	COLNVLDHVKWLRT-UHFFFAOYSA-N	2.54	166.086 [M+H] ⁺	166.086172
3	7-methoxy-9,10-dihydrophenanthrene-2,5-diol	C15H14O3	RDKDIPDDUFMMMT-UHFFFAOYSA-N	2.74	260.128 [M+H] ⁺	260.127638
4	Caffeine	C8H10N4O2	RYYVLZVUVIJVGH-UHFFFAOYSA-N	3.02	195.088 [M+H] ⁺	195.087512
5	Vasicinone	C11H10N2O2	SDIVYZXRQHWCKF-UHFFFAOYNA-N	3.14	203.081 [M+H] ⁺	203.081436
6	Cearoin	C14H12O4	NFJVELXCUBWAFL-UHFFFAOYSA-N	3.22	262.107 [M+H] ⁺	262.106666
7	Sarracenin	C11H14O5	QGBCGMGBGAHJIT-UHFFFAOYSA-N	3.77	227.091 [M+H] ⁺	227.090771
8	picein	C14H18O7	GOZCEKPECKLNO-RKQHYHRCSA-N	5.86	299.109 [M+H] ⁺	299.108993
9	Coumaroyl tyramine	C17H17NO3	RXGUTQNKCXHALN-UHFFFAOYSA-N	6.55	284.127 [M+H] ⁺	284.126979
10	Palmatine	C21H22NO4	QUCQEUCGKKTEBI-UHFFFAOYSA-N	6.63	352.153 [M+H] ⁺	352.153469
11	9-hydroxy-1,4a-dimethyl-7-propan-2-yl-2,3,4,9,10,10a-hexahydrophenanthrene-1-carboxylic acid	C20H28O3	GPFBVBJYXFRIOFB-UHFFFAOYSA-N	7.76	334.236 [M+H] ⁺	334.236389
12	N-(1-hydroxy-3-phenylpropan-2-yl) benzamide	C16H17NO2	RFYNNAVYPYXLVOM-UHFFFAOYSA-N	7.78	256.132 [M+H] ⁺	256.132418
13	Jasmonic Acid	C12H18O3	ZNJFBWYDHIGLCU-UHFFFAOYSA-N	7.87	211.132 [M+H] ⁺	211.132405
14	2-Cyclohexen-1-one, 4-hydroxy-4-(3-hydroxybutyl)-3,5,5-trimethyl-	C13H22O3	CWOFGGNDZOPNFG-UHFFFAOYSA-N	8.43	209.153 [M+H] ⁺	209.153089
15	2H-Cyclohepta[b]furan-2-one, 3,3a,4,7,8,8a-hexahydro-6-(3-hydroxybutyl)-7-methyl-3-methylene-	C15H22O3	KQEADODJYBGFC-UHFFFAOYSA-N	9.68	273.145 [M+H] ⁺	273.145146
16	3,8-dihydroxy-3,8-dimethyl-5-propan-2-ylidene-1,2,3a,4,7,8a-hexahydroazulen-6-one	C15H24O3	TXIKNNOOLCGADE-UHFFFAOYSA-N	9.83	253.179 [M+H] ⁺	253.17892
17	Resibufogenin	C24H32O4	ATLJNLJYJOCWJE-CWMZOUAVSA-N	10.01	385.236 [M+H] ⁺	385.235753
18	Pinoresinol dimethyl ether	C22H26O6	PEUUVVGQIVMSAW-RZTYQLBFSA-N	10.11	387.179 [M+H] ⁺	387.179251
19	(+)-Nootkatone	C15H22O	WTOYNNBCKUYIKC-JMSVASOKSA-N	11.87	219.174 [M+H] ⁺	219.173939
20	14-Deoxyandrographolide	C20H30O4	GVNRTWSGBWPJGS-YDSKTCISA-N	12.08	335.218 [M+H] ⁺	335.21831
21	Lindenol	C15H18O2	XRDJYSVGPBJZSG-PSDLAXTLISA-N	12.43	231.137 [M+H] ⁺	231.137161
22	Celastrol	C29H38O4	KQJSQWZMSAGSHN-JJWQIEBTSA-N	12.83	473.265 [M+H] ⁺	473.265247
23	dehydroabietic acid	C20H28O2	NFWKVWVWBFBAOV-IPNZSQQUA-N	13.95	318.241 [M+H] ⁺	318.241489
24	Naringenin chalcone	C15H12O5	YQHMWTPYORBCMF-ZZXKWWIFSA-N	0.89	271.058 [M-H] ⁻	271.05809
25	Sucrose	C12H22O11	CZMRCDWAGMREC-UGDNZRGBSA-N	1.25	341.109 [M-H] ⁻	341.108786
26	PYROCATECHUIC ACID	C7H6O4	GLDQAMYCGOJJDV-UHFFFAOYSA-N	1.74	153.019 [M-H] ⁻	153.019331
27	Cinnamic acid	C9H8O2	WBWYXWJHAXSJN-VOTSOKGWSA-N	4.31	193.050 [M-H] ⁻	193.050275
28	Paracetamol	C8H9NO2	RZVAJINKPMORJF-UHFFFAOYSA-N	4.36	150.056 [M-H] ⁻	150.056008
29	Suberic acid	C8H14O4	TYFQFVWCELRYAO-UHFFFAOYSA-N	5.03	173.082 [M-H] ⁻	173.081768
30	Apigenin-7-O-glucoside	C21H20O10	KMOUJOKENFFTPU-QNDFHXLGSA-N	5.84	431.098 [M-H] ⁻	431.098137
31	3,4,5-trimethoxycinnamic acid	C12H14O5	YTFVRYKNXDADBI-SNAWJCMRSA-N	6.47	237.077 [M-H] ⁻	237.076615
32	Ethyl ferulate	C12H14O4	ATJVZXXHKSVELS-FNORWQNLISA-N	7.50	221.082 [M-H] ⁻	221.081508
33	Polygalic acid	C29H44O6	VZRKWGPZJDNHC-UHFFFAOYSA-N	10.80	487.306 [M-H] ⁻	487.306153
34	(2R,7S,13R,14R,16S,19R,20S)-19-(furan-3-yl)-11-hydroxy-9,9,13,20-tetramethyl-4,8,15,18-tetraoxahexacyclo[11.9.0.0 ^{2,7} .0 ^{2,10} .0 ^{14,16} .0 ^{14,20}]docos-10-ene-5,12,17-trione	C26H28O9	SNGLHUWTFLYPMT-OEZBNQCDSA-N	10.97	483.163 [M-H] ⁻	483.162505
35	Spiculisporic acid	C17H28O6	TUXHHVJPGQUPCF-DYVFJYSZSA-N	11.19	327.181 [M-H] ⁻	327.181137
36	Palmitic acid	C16H32O2	IPCSVZSSVZVIGE-UHFFFAOYSA-N	17.79	255.232 [M-H] ⁻	255.232385

RT: retention time.

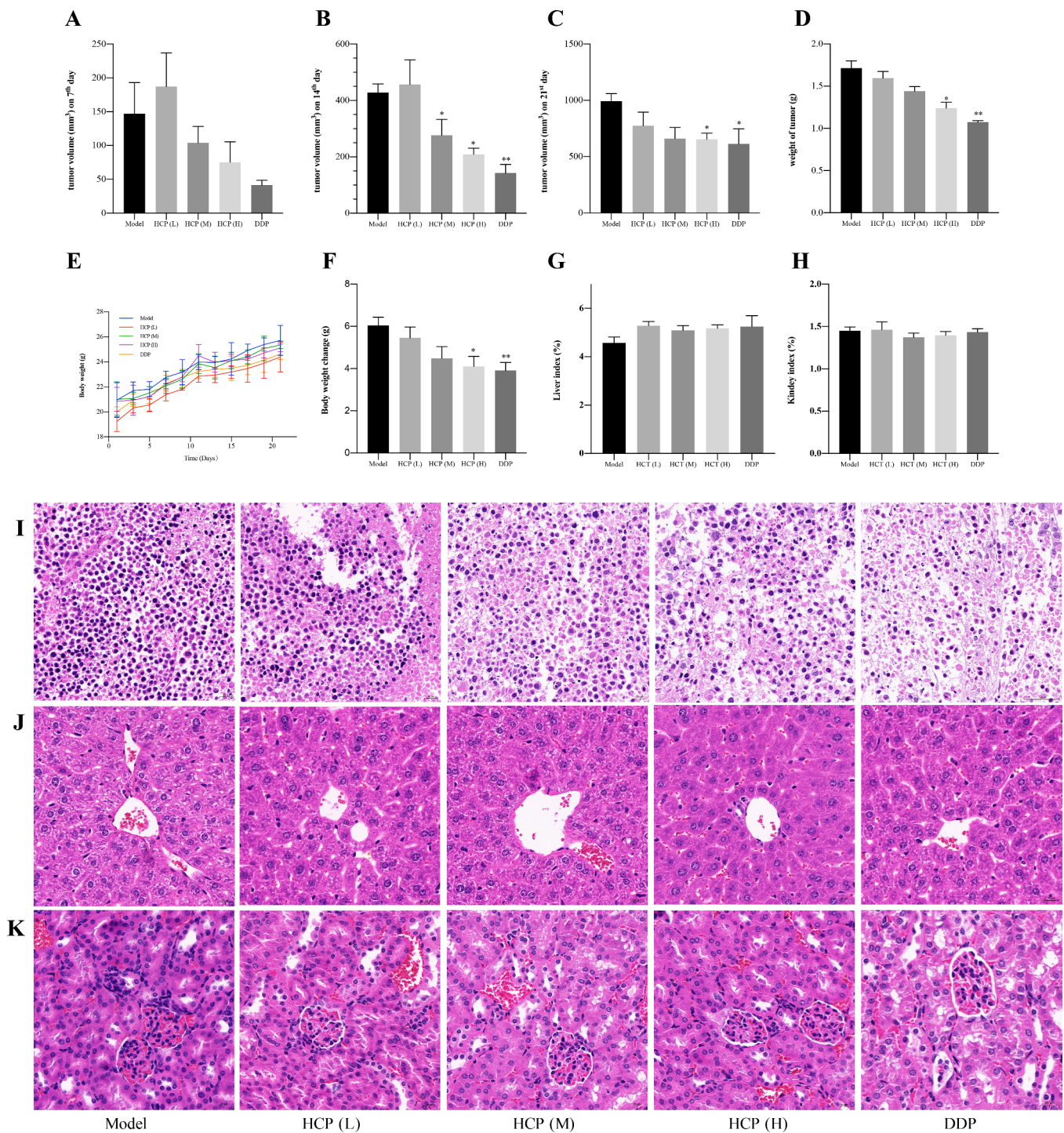


Fig. 2. HCP inhibits lung cancer tumor growth *in vitro* and has no toxicity on the liver and kidney. (A) Tumor volume (mm³) on day 7. (B) Tumor volume (mm³) on day 14. (C) Tumor volume (mm³) on day 21. (D) Tumor weight (g) on day 21. (E) Body weights (g) measured every 3 days. (F) Body weight (g) change. (G) Ratio of liver weight to body weight. (H) Ratio of kidney weight to body weight. (I) Tumor tissue slides were stained with H&E. (J) Liver tissue slides were stained with H&E. (K) Kidney tissue slides were stained with H&E. The scale bar represents 20 μ m. * $p < 0.05$ and ** $p < 0.01$ compared with the model group. All data are presented as means $\pm$ SD ($n = 8$).

H&E, cancer cells were large and heteromorphous, with large and intensely stained nuclei. The tumor cells were closely arranged, with many large nuclei and clear morphologies in the model group. In the DDP group, the distribution of tumor cells was relatively loose, with relatively few cells. Compared with the model group, the HCP-treated

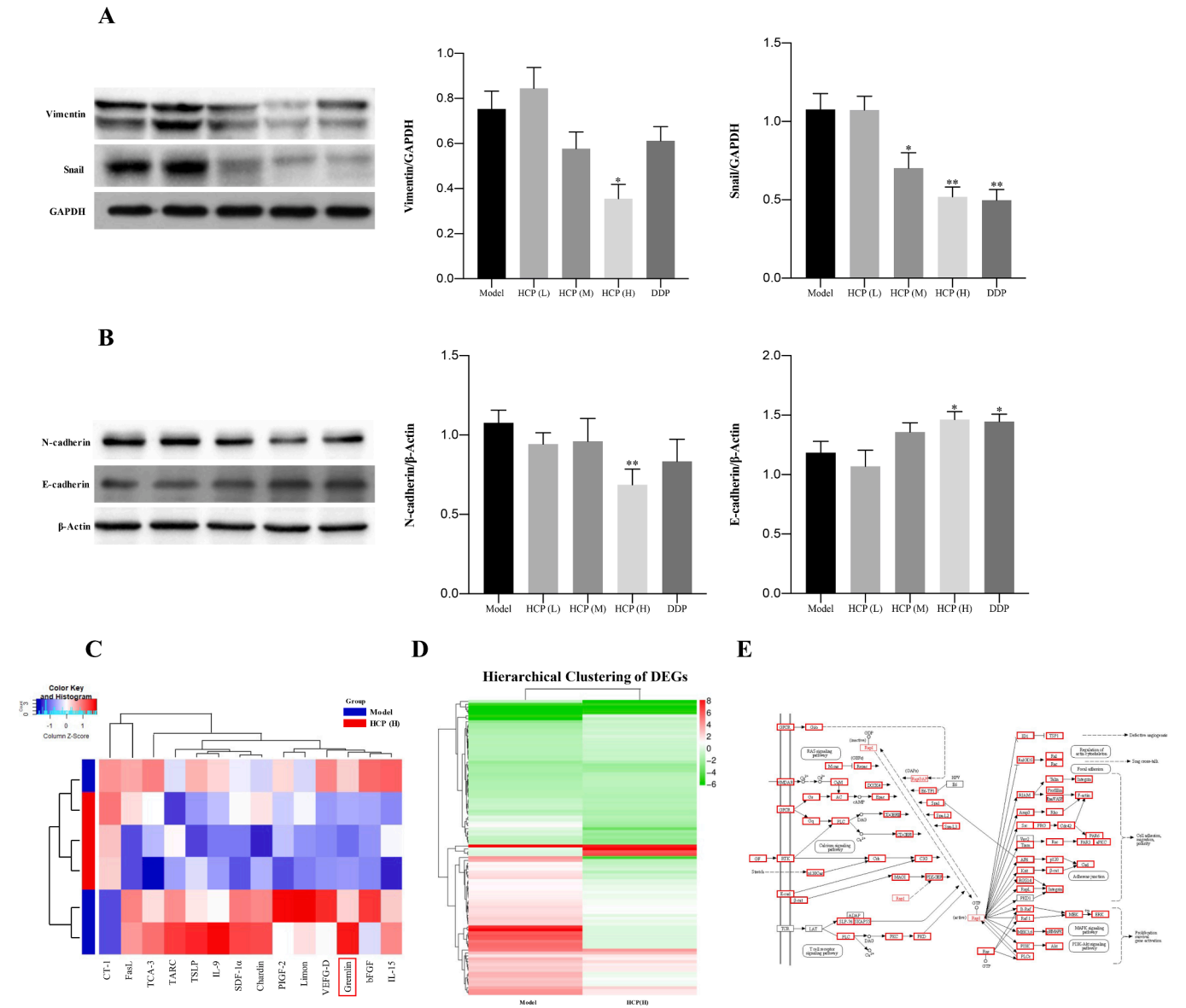
groups showed relatively few abnormal cells and loose cell densities (Fig. 2I). Liver tissue sections of each group were observed. The liver lobules of each group of mice were intact, the liver cells were not enlarged, the nuclei were positioned in the middle, the intercellular spaces were clear, and there were no evident abnormalities (Fig. 2J). No

significant pathological changes were found in the renal tissue sections in each group. There were no increases in the number of membrane cells or endothelial cells in the glomerulus, inflammatory exudate, or necrosis or other abnormalities such as degeneration, atrophy, hyperplasia, or expansion of the renal tubules (Fig. 2K).

HCP inhibits EMT, GREM1 level, and associated pathways

Western blotting indicated that the levels of vimentin and N-cadherin were significantly lower in the HCP (H) group than in the model group. Snail level was significantly decreased in the HCP (M), HCP (H), and DDP groups. Additionally, E-cadherin levels were increased in the HCP (H) and DDP groups (Fig. 3A, B). We found that HCP (H) had the best inhibitory effect on tumor growth compared to those of other treatments based on animal experiments. Thereafter, we analyzed the protein chip data for the comparison between the model group and the HCP (H) group. This suggested that 14 proteins showed differential

expression (4 downregulated and 10 upregulated) (Fig. 3C). One of the upregulated proteins was GREM1, which is closely related to cancer metastasis and is encoded by the *GREM1* gene (Chen et al., 2019). Furthermore, a comparison of samples in the HCP(H) and model groups using high-throughput sequencing revealed 1121 differentially expressed genes (441 downregulated and 680 upregulated) (Fig. 3D). Using multiple database (miRanda, RNAhybrid, mirDIP, and TargetScan), we performed a target prediction analysis of the differentially expressed miRNAs obtained using sequencing. Many genes were involved in signaling pathways, with enrichment for the Rap1 signaling pathway. This signaling pathway may mediate the effects of HCP; accordingly, we focused on the Rap1 signaling pathway. Further analysis revealed that *GREM1* expression is regulated by miR-205-5p (Fig. 3E). Therefore, we proposed that HCP exerts antitumor effects by the regulation of the Rap1 signaling pathway and that GREM1 is an important therapeutic target for the metastasis of NSCLC.



GREM1 promotes metastasis by regulating the EMT and Rap1 pathway

The effect of GREM1 on proliferation in A549 cells was evaluated using a CCK-8 assay. OE-GREM1 in A549 cells did not significantly inhibit or promote proliferation (Fig. 4A). When we transfected A549 cells with shGREM1, the same result was observed (Fig. 5A). The effect of GREM1 on metastasis was explored using wound healing and Transwell assays. OE-GREM1 significantly increased the migration and invasion of A549 cells (Fig. 4B, C). Furthermore, when A549 cells were transfected with shGREM1, migration and invasion were markedly reduced (Fig. 5B, C). These data confirmed that GREM1 promotes the metastasis of NSCLC cells and does not significantly increase cell proliferation.

To further explore the mechanism underlying the effect of GREM1 on metastasis, we studied its role in EMT. As GREM1 expression increased, the protein levels of mesenchymal markers (including vimentin, Snail, and N-cadherin) increased significantly. By contrast, the level of the epithelial marker E-cadherin decreased dramatically (Fig. 4D). The expression of mesenchymal markers was inhibited, and epithelial marker expression increased significantly in A549 cells treated with shGREM1 (Fig. 5D). Therefore, GREM1 might promote migration and invasion in NSCLC by promoting EMT progression. We further investigated changes in Rap1-associated signaling pathway components (Rap1, c-raf, and erk1/2) in response to GREM1 in NSCLC. We found that GREM1 overexpression significantly increased expression of the signaling protein-related genes in the Rap1 pathway (Fig. 4D) and vice versa (Fig. 5D). Therefore, these results revealed that GREM1 plays an important role in the regulation of the Rap1 pathway and might promote metastasis in NSCLC by activating this pathway.

GREM1 does not promote proliferation in A549 xenograft mouse models

To examine the role of GREM1 in the growth of A549 cells *in vivo*, we performed xenograft tumor assays using nude mice (Fig. 6A–B). The body weights of mice with A549 cells stably transfected with OE-GREM1 were not significantly different from those of the other groups (Fig. 6C). However, on days 42, 46, and 48, the tumor volume and tumor growth in the OE-GREM1 group with HCP treatment were lower than those in the other groups. On day 48, the tumor volume and weight in this group were significantly lower than those in the control group (Fig. 6D,E). These data suggested that GREM1 did not promote A549 cell growth *in vivo*, although HCP had an inhibitory effect.

MiR-205-5P directly regulates GREM1 expression

To verify whether miR-205-5P interacts with GREM1 and inhibits its expression, we constructed plasmids expressing WT and mutant GREM1. The dual-luciferase reporter system was used to determine GREM1 expression. The fluorescence ratio of GREM1 co-transfected with miR-205-5p mimics was significantly reduced (Fig. 6F, G). These findings indicated that GREM1 promotes NSCLC metastasis, and its expression is negatively regulated by miR-205-5P.

Discussion

Lung cancer is one of the deadliest malignant tumors worldwide. Surgery is the conventional treatment for early-stage lung cancer, whereas chemotherapy, radiotherapy, targeted therapy, and immunotherapy are available for patients with advanced disease (Xu et al., 2021). These therapies produce toxic side effects, which are intolerable to many patients, limiting their general acceptance. TCM enables the treatment of lung cancer with relatively few side effects, improves clinical symptoms, prolongs survival, and improves immunity in patients (Wang et al., 2020). It has played an increasingly important role in the treatment of lung cancer owing to its unique advantages. The principles of TCM in disease treatment are “person-centered,” “treating both

symptoms and root causes,” “overall regulation,” and “treatment based on syndrome differentiation.” According to Professor Zhou, lung cancer involves various pathological factors, such as “blood stasis,” “phlegm,” “deficiency,” and “toxicity.” Treatment focuses on “abundant qi,” including “nourishing lung and strengthening the spleen and stomach” and “strengthening the essence and nourishing the kidney.” Therefore, the goal of TCM for the treatment of lung cancer is to adjust the balance of qi, blood, yin, and yang in the body, to confer a state of “Yin-Yang” balance, and to eliminate pathogens, e.g., by dissipating phlegm, removing blood stasis, and detoxification (Zhou, 2011). Professor Zhou followed the method of “invigorating qi and removing phlegm” and developed HCP, which has been used widely for lung cancer treatment for over 50 years, particularly for advanced lung cancer. Previous clinical studies have shown that HCP can inhibit lung cancer progression, alleviate clinical symptoms, improve immunity, and prolong survival without toxic side effects (Zhou et al., 1986). HCP has been combined with chemotherapy to reduce the incidence of side effects of chemotherapy and prolong survival (Lu and Chen, 2006). Research has demonstrated that HCP promotes apoptosis and exerts an antitumor effect (Xiong et al., 2020; Zhu et al., 2006). Our first animal experiment provided evidence that HCP could effectively inhibit tumor growth in mice.

Owing to the complex and diverse composition of Chinese herbal compounds, it is difficult to elucidate the mechanism underlying the antitumor effect of HCP. Based on the results of UPLC-MS, ten monomers had antitumor effects. Cearoin can induce various forms of cell death in neuroblastoma SH-SY5Y cells, including autophagy, ERK activation, and apoptosis, and its effects are mainly mediated by reactive oxygen species (ROS) production, supporting its application for the treatment of neuroblastomas (Ju et al., 2019). Sarracenin is an active component of *Patrinia heterophylla*, which is effective against tumors (Sheng et al., 2019). Cytotoxic activity of picein against the human lung cancer cell line H1299 has been reported (Jeon et al., 2008). Resibufogenin is a potential antitumor agent for the treatment of NSCLC, and its effects may be triggered by Caspase-1-dependent pyroptosis via the ROS-mediated inhibition of NF- κ B (Yin et al., 2021). The amount of 14-deoxyandrographolide extract from andrographolide is greater than that of *Andrographis paniculata*, and its cytotoxic effect on cancer cells is 4-fold greater (Yin et al., 2021). Celastrol has been identified as an effective drug for triple-negative breast cancer, involving a reduction in IL1 β expression, activity, or signaling pathways (You et al., 2021). New dehydroabietic acid derivatives with anticancer activity against HepG2 cell lines are potential drugs targeting the kinase domain of EGFR (Zaki et al., 2021). Cinnamic acid is an antitumor agent that can be used alone or with conventional chemotherapeutic agents, such as cisplatin, and its effects could be attributed to its proapoptotic activity (Almeer et al., 2019). Apigenin-7-O-glucoside, a bioactive compound derived from natural plants, inhibits proliferation and induces apoptosis in cervical cancer cells via ROS production and G0/G1 phase arrest (Liu et al., 2020). Palmitic acid is a powerful inhibitor of the IL-10-STAT3-NF- κ B signaling axis and EMT in colorectal cancer cells (de Araujo Junior et al., 2020). These components/metabolites of HCP have an inhibitory effect on tumors, providing a basis for exploring the molecular mechanism underlying the antitumor activity of HCP. However, the main components of HCP that inhibit metastasis remain unclear and need to be explored.

Invasion and metastasis are key cancer characteristics resulting in clinical treatment failure and recurrence; therefore, the inhibition of these processes is a focus of new antitumor drug development (Wang et al., 2013). Therefore, it is necessary to clarify the mechanism underlying the effects of HCP on lung cancer, as this information is expected to provide new directions for diagnosis and treatment. Recurrence and metastasis are the major causes of death in patients with lung cancer. The occurrence of lung cancer is a complex process involving the synergistic action of multiple genes. This process is closely related to the activation of oncogenes and the inactivation of tumor

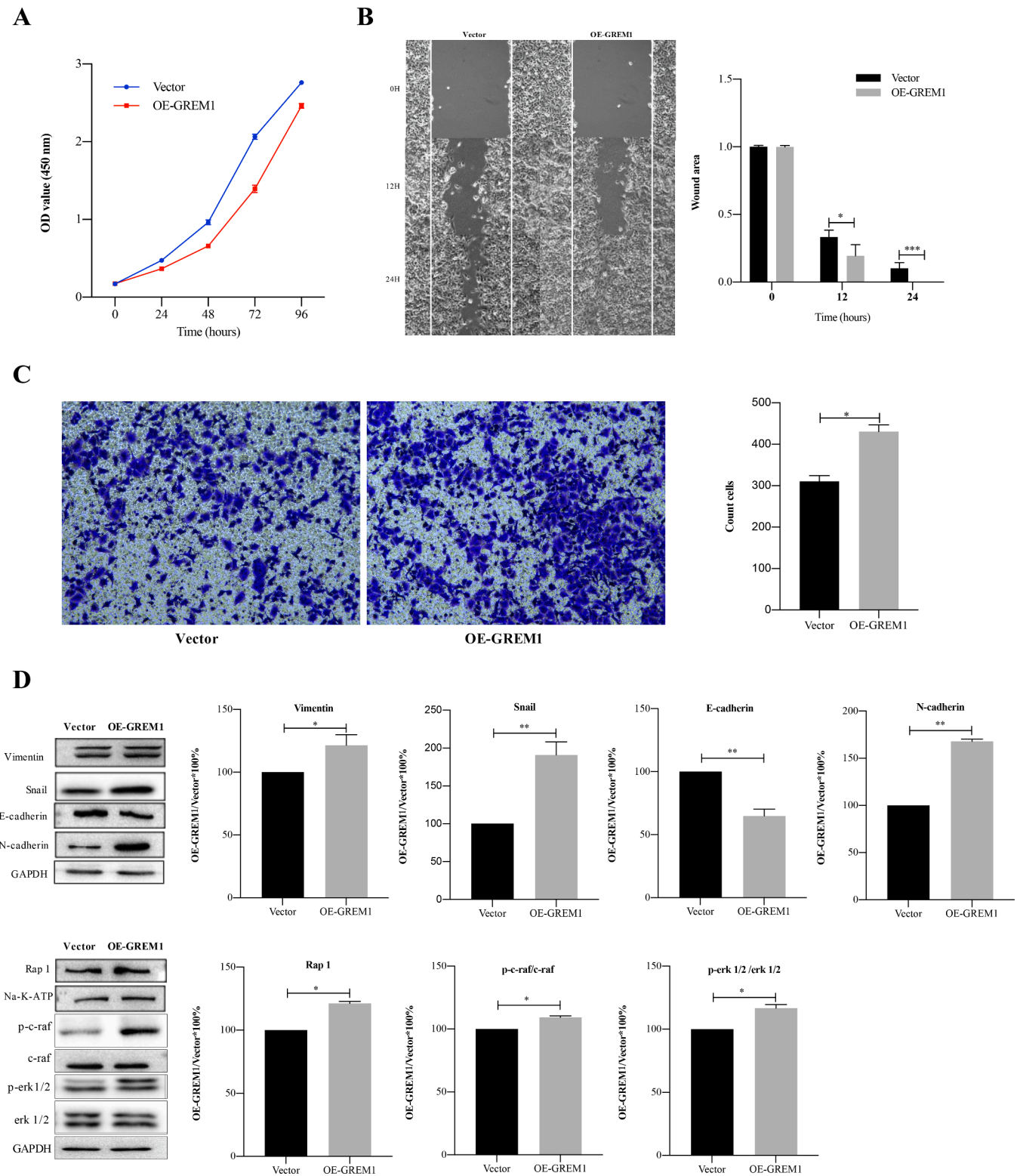


Fig. 4. OE-GREM1 promotes A549 cell metastasis by regulating EMT progression via the Rap1 pathway. (A) Effects of OE-GREM1 on the proliferation of A549 cells, as determined using a CCK-8 assay. (B) Representative images (100 × magnification) of OE-GREM1 cell migration at 0, 12, and 24 h after scratch wounding and the quantification of results. White lines identify margins. (C) Transwell assay showing that OE-GREM1 promotes the invasion of A549 cells and quantification of results. (D) The protein expression of OE-GREM1 in A549 cells. The expression levels of GAPDH and Na-K-ATP were used as internal references. Quantitative analysis of protein expression. Data are presented as means ± SD (N = 3). * $p < 0.05$ and ** $p < 0.01$ compared with the vector group.

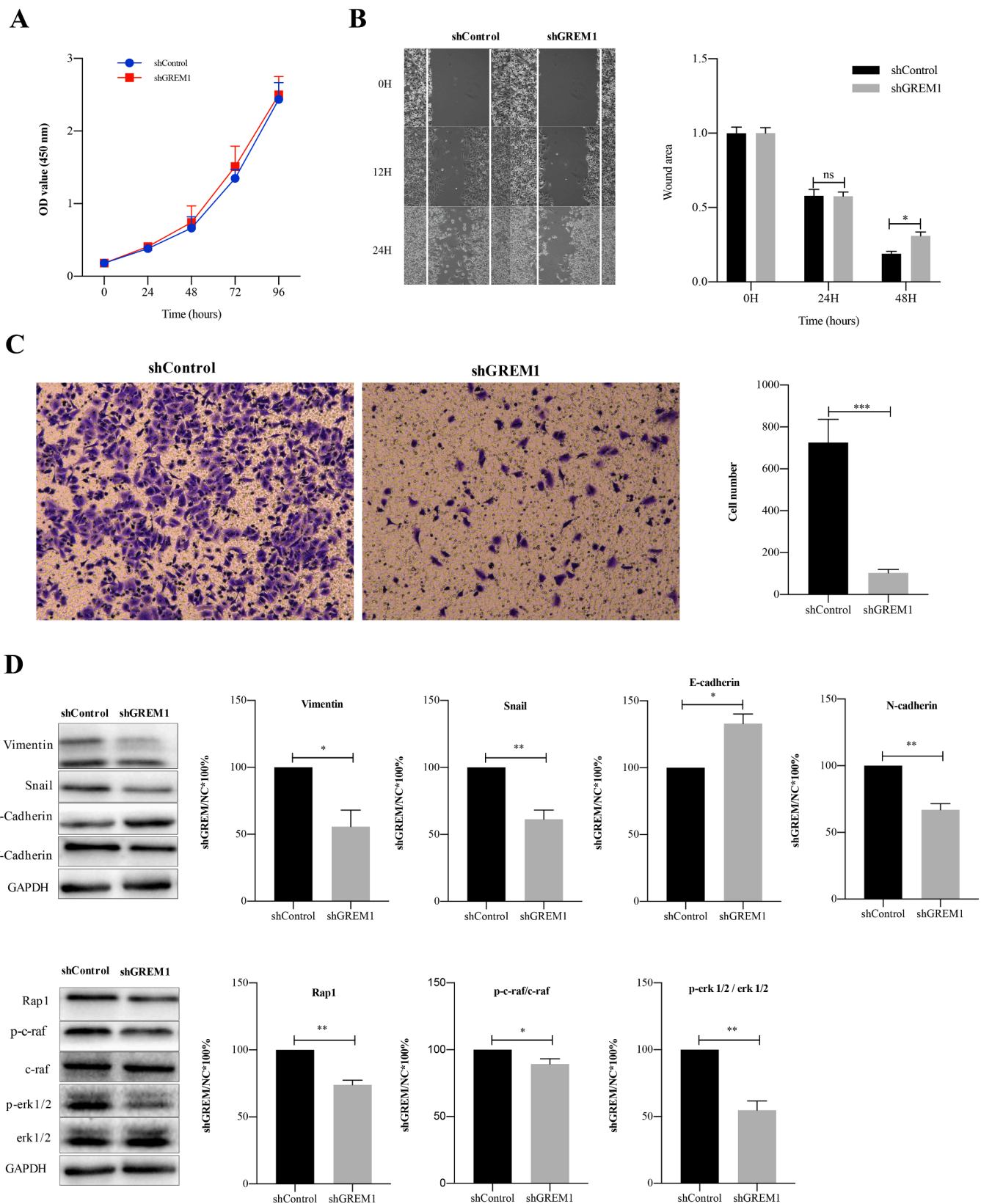


Fig. 5. ShGREM1 affected metastasis of A549 cells by the regulation of EMT progression via the Rap1 pathway. (A) Effects of shGREM1 on the proliferation of A549 cells, as determined using a CCK-8 assay. (B) Representative images (100 × magnification) of shGREM1 cell migration at 0, 12, and 24 h after scratch wounding and the quantification of results. White lines identify margins. (C) Transwell assay showing that shGREM1 inhibits the invasion of A549 cells and quantification of results. (D) Protein expression of shGREM1 in A549 cells. The expression of GAPDH was used as an internal reference. Quantitative analysis of protein expression. Data are presented as means ± SD (N = 3). * $p < 0.05$ and ** $p < 0.01$ compared with the shControl group.

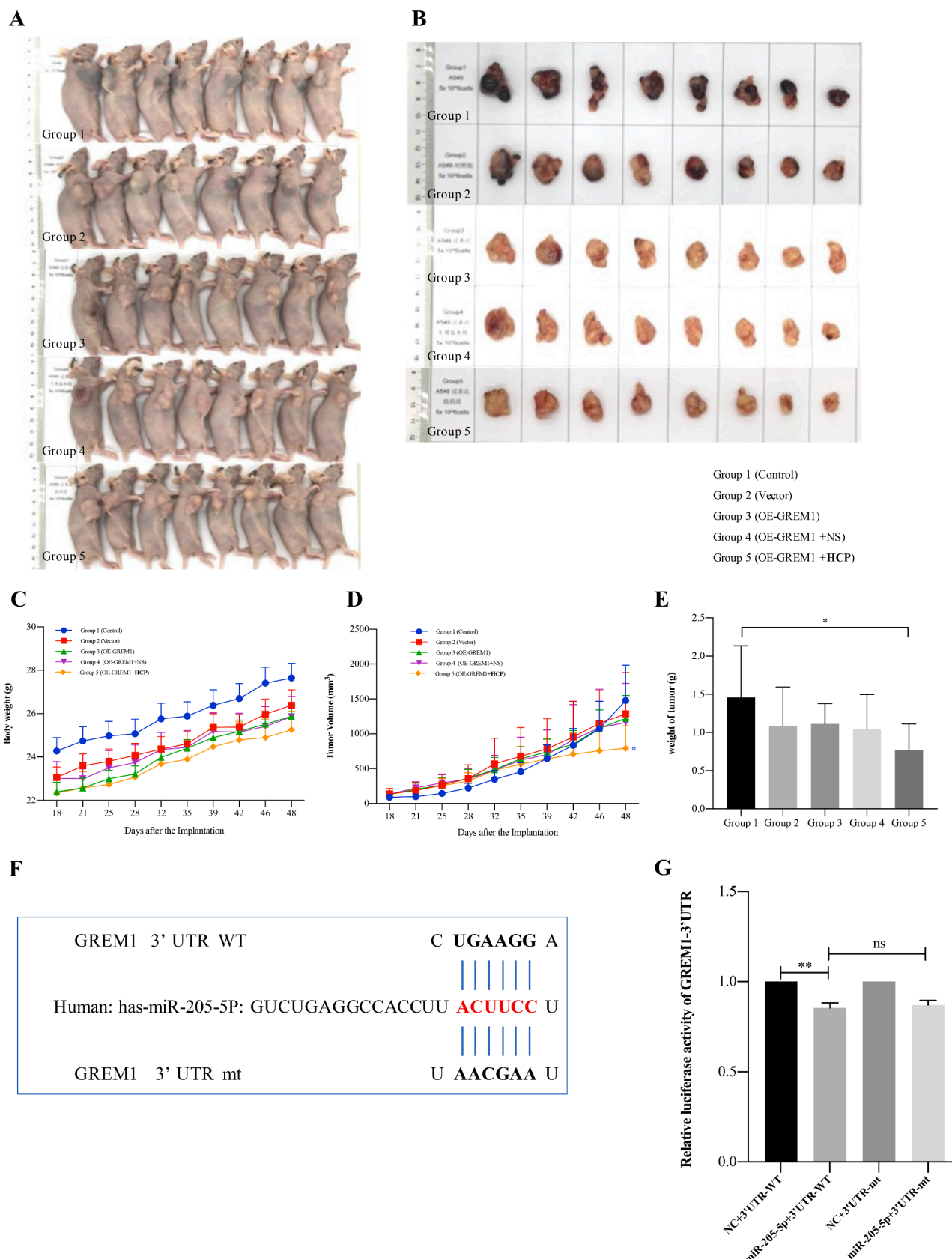


Fig. 6. OE-GREM1 does not affect the proliferation in A549 xenograft nude mouse models, and miR-205-5P directly regulates GREM1 expression. (A) Photographs of mice bearing human A549 xenografts. (B) Images of excised tumors. (C) Body weights measured twice a week. (D) Growth curves for xenograft tumors. (E) Tumor weight (g). (F) Schematic illustration of the hypothetical duplexes formed by miR-205-5P with the 3' UTR of *GREM1*. (G) Luciferase activities as quantified from 293T co-transfected with the WT or mutant 3' UTR of *GREM1* luciferase reporter plasmids together with the miR-205-5p mimic or the corresponding control. Data are presented as means $\pm$ SD ($N = 8$). * $p < 0.05$ and ** $p < 0.01$.

suppressor genes (Petpiroon et al., 2019). Cell proliferation, loss of cellular adhesion, ECM degradation, and cell migration and invasion lead to tumor cell metastasis and recurrence (Wang et al., 2013). Tumor metastasis depends on the ability of cells to compete with normal cells in the *in vivo* environment (Harries et al., 2018). In our study, western blotting was used to detect the levels of EMT markers in tumor tissues, revealing that HCP could inhibit tumor growth in mice by suppressing EMT. Protein chips are a new high-throughput technology used to evaluate changes in gene and protein expression levels in tumors, and this approach could be used to identify new candidates for clinical targeted therapy (Zhao et al., 2020).

Fourteen differentially expressed proteins were identified in our study. Compared with the levels in the HCP(H) group, levels of 10 proteins were upregulated in the model group: FasL, TCA-3, TARC, TSLP, IL-9, SDF-1 α , Chardin, Limon, Gremlin, and IL-15. Fas ligand (FasL) is a transmembrane protein belonging to the tumor necrosis factor family that induces apoptosis in target cells by binding to its receptor Fas, containing the death domain (de Jesus et al., 2016). TCA-3 is a mouse glycoprotein belonging to the β -chemokine family, which can inhibit tumor growth *in vivo* by inducing lymphocyte-independent antitumor activity and stimulating tumor-specific immunity (Oliveira et al., 2002). Thymus and activation-regulated chemokine (TARC) is a Th2-type driver that is constitutionally expressed in the thymus and is produced by a variety of different cells (Tepljakov et al., 2018). TARC is closely correlated with inflammation and tumor occurrence. Systemic inflammation induced by tumor inflammatory cells can prompt transformation into a tumor-inflamed internal environment, thereby causing a series of tumors to develop, progress, and metastasize (Komatsu-Fujii et al., 2018; Liu et al., 2019). In the inflammatory environment, tumor cells can damage vascular endothelial cells and activate platelets to release vascular endothelial growth factors to repair endothelial cells, promoting the formation of novel tumor blood vessels. As a result, tumor cells adhere to new blood vessels, leading to a series of side effects, such as tumor proliferation, migration, and distant metastasis (Franco et al., 2015). TARC chemokines promote inflammatory responses that indirectly lead to tumor formation. Interleukin (IL)-9, IL-15, and thymic stromal lymphopoietin (TSLP) are members of the common gamma chain family of cytokines and pleiotropic cytokines and have various effects on a variety of cells and tissues (Jia et al., 2017; Yamaguchi et al., 2020). Stromal cell-derived factor-1 alpha (SDF-1 α) is a chemokine and chemokine receptor with a vital role in tumorigenesis, including tumor cell proliferation, metastasis, and angiogenesis (Peng et al., 2016). Gremlin is a highly conserved secreted protein and is usually overexpressed in gastrointestinal malignant tumors; it can promote tumor invasion and metastasis and induce EMT and migration (Binhui et al., 2018; Chen et al., 2019). However, there are few reports on the relationship between GREM1 and lung cancer. GREM1 was identified by the protein chip analysis as highly expressed in the model group, suggesting that its high expression may be closely related to invasion and metastasis in NSCLC. This study provides an entry point for further research.

The EMT is a transformation process by which epithelial cells gain a mesenchymal phenotype after a specific program. Tumor cells often obtain mesenchymal characteristics by EMT and have high migration, invasion, and ECM degradation abilities. Therefore, the promotion of EMT is crucial for the processes of tumor invasion and metastasis (Guo et al., 2020). In metastasis, the main manifestation of EMT in tumor cells is the downregulation of expression of a specific marker of epithelial cells, E-cadherin, and the upregulation of expression of the mesenchymal cell markers vimentin and N-cadherin (Ying et al., 2018). A low expression level of E-cadherin can directly weaken interactions between tumor cells, making it easier for tumor cells to detach from tissues, and therefore, this factor is a key node in the EMT (Zou et al., 2013). The transcription factor Snail plays a vital regulatory role in the EMT. The regulation of Snail is a prerequisite for tumor cell invasion. It can affect or regulate the differential expression of E-cadherin, vimentin, and N-cadherin (Greco et al., 2016). GREM1 overexpression significantly

altered the protein levels of epithelial and mesenchymal markers in lung cancer cells. These results provide strong evidence that GREM1 plays a vital role in initiating the EMT process and may play a role in promoting cell invasion and migration.

Rap1 is a small GTPase belonging to the Ras family, marking the malignant expression of fibroblasts transformed by KRAS (Vlasschaert et al., 2015). The Ras family regulates various benign and malignant biological behaviors, such as cell adhesion, growth, apoptosis, motility, and activated Ras gene mutations, leading to the occurrence of malignant tumors (Maertens and Cichowski, 2014). High expression of Rap1 can impair cell adhesion or enhance cell-matrix adhesion, which can increase tumor cell migration and invasion (Tsygankova et al., 2010). The Raf/Erk signaling pathway functions downstream of Rap1. The activation of Rap1 via growth factors regulates Erk signal transduction, resulting in cell proliferation and migration (Han et al., 2020). The Raf/Erk signaling pathway is involved in tumorigenesis and metastasis and is the main pathway for cell growth, proliferation, and differentiation. Erk1/2 is the most common protein in the MAPK cascade and plays a critical role in regulating cell growth and differentiation in the Raf/Erk signaling pathway. The tumor-promoting gene *Bcl-2* is the downstream substrate of Erk1/2. After Ras protein expression is upregulated, it can activate the Raf/Erk signaling pathway, which leads to an increase in *Bcl-2* expression and the immortalization of tumor cells (Xiao et al., 2015). We speculated that GREM1 has a potential regulatory role in the Rap1 pathway. The expression of GREM1 was changed, and the expression levels of Rap1, p-c-Raf, and p-erk1/2 in the Rap1 pathway also changed significantly. Therefore, GREM1 might enhance the metastasis of NSCLC cells via the Rap1 pathway. miRNAs are short non-coding single-stranded RNAs. They have regulatory effects on transcription and mainly regulate a series of events, such as translation, degradation, and expression of target mRNAs (Correia de Sousa et al., 2019). miRNAs participate in all important biological processes, such as embryonic development, tumor occurrence, progression, invasion, and metastasis, apoptosis, death, and autophagy (Chen et al., 2019). Our luciferase activity indicated that GREM1 is a direct target of miR-205-5p. Moreover, xenograft experiments confirmed that OE-GREM1 does not influence cell proliferation. However, the tumor volume after HCP intervention was reduced significantly, clearly establishing the inhibitory effect of HCP on tumors.

Conclusion

Thirty-six ingredients were classified from the serum of HCP-treated mice. HCP can slow the growth rate of subcutaneously transplanted tumors in nude mice to a certain extent. HCP had no effect on the liver and kidney in a lung cancer mouse model. No side effects were observed. The mode of action of HCP may be related to its inhibitory effect on NSCLC metastasis via the Gremlin/Rap1 signaling pathway regulated by miR-205-5p.

CRedit authorship contribution statement

Jun Kan: Formal analysis, Visualization, Conceptualization, Writing – original draft. **Biqian Fu:** Formal analysis, Visualization, Conceptualization, Writing – original draft. **Ruisheng Zhou:** Visualization. **Daihan Zhou:** Visualization. **Yufang Huang:** Formal analysis. **Hongwei Zhao:** Formal analysis. **Yunlong Zhang:** Conceptualization. **Yuming Rong:** Formal analysis. **Jun Dong:** Formal analysis. **Liangping Xia:** Formal analysis. **Shanshan Liu:** Formal analysis. **Qiuling Huang:** Formal analysis. **Nannan Wang:** Formal analysis. **Na Ning:** . **Bei Zhang:** Writing – review & editing. **Enxin Zhang:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no competing financial interests. All data were generated inhouse, and the manuscript was written by the

authors. All authors agree to be accountable for all aspects of work ensuring integrity and accuracy.

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